

SALES PERFORMANCE DASHBOARD

Submitted by: JESMAA E

CERTIFICATE

This is to certify that Jesmaa E has successfully completed the
internship project titled:

“Power BI Sales Performance Dashboard”

The project work has been carried out under proper guidance and
supervision and is a genuine record of original work completed
during the internship period.

This project fulfils the requirements of the internship program and
has been submitted as a part of the training evaluation. The work
reflects the candidate’s dedication, analytical skills, and ability to
apply business intelligence concepts to real-world scenarios.

Signature & Seal

(Team Lead / Internship Mentor)

DECLARATION

I, **Jesmaa E**, hereby declare that the project titled:

“Power BI Sales Performance Dashboard “

is an original work carried out by me during my internship. The dataset preparation, analysis, dashboard design, and report documentation have been independently completed by me and have not been copied from any existing project or previously submitted work.

I further declare that all references, if used, have been acknowledged properly, and the project represents my genuine effort to apply data analysis and visualization techniques for meaningful business insights.

Date: 21-12-2025

Jesmaa E

ACKNOWLEDGMENT

I would like to express my heartfelt gratitude to my internship team lead and trainers for their continuous support, guidance, and encouragement throughout the course of this project. Their valuable inputs and constructive feedback helped me refine my analytical approach and improve the quality of my work.

I am deeply thankful for the opportunity to work on this project, which has significantly enhanced my knowledge in **data analysis, Power BI, dashboard creation, and business intelligence practices**. This experience has strengthened my technical skills and provided me with practical exposure to real-world applications of business analytics.

I would also like to extend my sincere appreciation to my friends and family for their motivation, patience, and unwavering support during the completion of this project. Their encouragement kept me focused and determined to deliver my best.

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Abstract

The main objective of this project is to analyze sales performance using Power BI and derive meaningful business insights. This project focuses on identifying key sales trends, category-wise performance, monthly variation in sales, profitable regions, and product demand behavior.

Beyond descriptive analytics, the project emphasizes **diagnostic analysis** (why sales fluctuate), **predictive insights** (seasonal demand forecasting), and **prescriptive recommendations** (strategic actions for management). The interactive dashboard helps business leaders make data-driven decisions, improve profitability, and plan better sales strategies.

Chapter 1: Introduction

1.1 Scope of Analysis

The analysis focuses on understanding:

- Overall Sales Performance
- Profitability Trends
- Category & Segment-wise Sales
- Regional Performance
- Product Demand
- Monthly Sales Trend Patterns

Additionally, the scope extends to:

- Identifying **top-performing products** and **low-performing categories**.
- Evaluating **customer segments** to understand buying behavior.
- Assessing **seasonality** and its impact on inventory planning.
- Providing **strategic recommendations** for marketing and supply chain optimization.

1.2 Approach of Analysis

- 1] Clean raw dataset using Python / Power BI

- 2 Load cleaned data into Power BI
- 3 Perform data modeling & field setup
- 4 Create dashboards using KPIs, charts & maps
- 5 Analyze and interpret business results

This structured analytical approach ensures accuracy and meaningful insights. It combines **ETL (Extract, Transform, Load)** processes with **visual storytelling**, making the dashboard not just a reporting tool but a **decision-support system**.

Chapter 2: Gathering Data

2.1 Dataset Overview

The dataset used for this project contains:

- Customer Information
- Order & Product Details
- Sales Value
- Profit & Quantity Sold
- Region, State & City Details

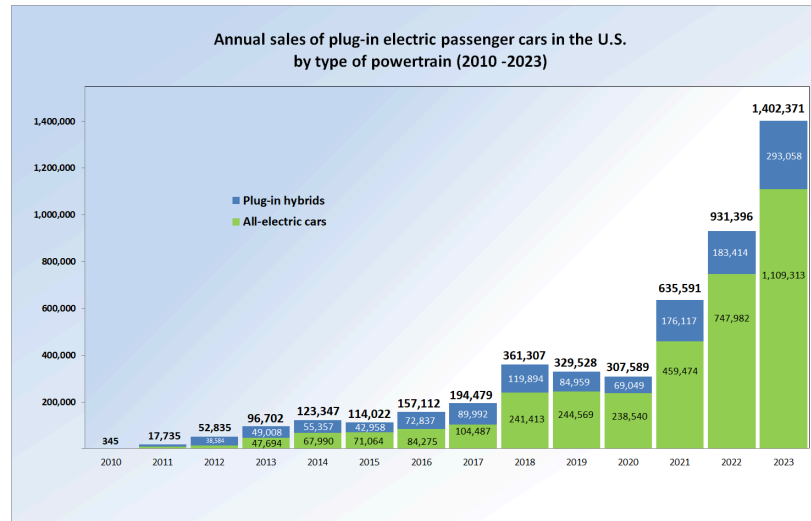
It spans multiple years, allowing **trend analysis over time**. The dataset is transactional in nature, meaning each row represents a unique order, which makes it suitable for **granular analysis**.

2.2 Data Structure

Important dataset fields include:

- **Order ID** – Unique sales transaction
- **Product Name** – Item purchased
- **Category** – Technology, Furniture, Office Supplies
- **Sales** – Revenue Generated
- **Quantity** – Units Sold
- **Profit** – Net Profit Earned
- **Country, State, City, Region** – Geographical fields

This structure allows multi-dimensional analysis, enabling slicing and dicing by **time**, **geography**, **product**, and **customer segment**.



Chapter 3: Data Preparation & Exploration

3.1 Data Cleaning

Performed steps:

- Removed blank & duplicate records
- Checked null values
- Verified numerical values
- Standardized column names

Additional steps included:

- Creating **calculated fields** (e.g., Profit Margin %).
- Handling **outliers** in sales values.
- Ensuring **date formats** were consistent for time-series analysis.

3.2 Exploratory Data Analysis

Performed checks to:

- Identify **top-selling products** and categories.
- Detect **most profitable regions**.
- Observe **monthly demand variations**.
- Evaluate **segment contribution**.

Exploration also revealed:

- Certain **products drive disproportionate revenue**.
- **Seasonal spikes** occur during festive months.
- **Regional preferences** differ (e.g., furniture sells more in urban regions).



The dataset was cleaned in **Google Colab** using Python libraries like *Pandas* and *NumPy*. Duplicate records were removed, missing

values handled, column names standardized, and numerical fields validated to ensure accuracy. This preprocessing made the data consistent, reliable, and ready for visualization in Power BI.

Chapter 4: Business Intelligence Dashboard

4.1 Dashboard Components

The dashboard contains:

- **KPI Cards**
 - Total Sales
 - Total Profit
 - Total Orders
 - Total Quantity
- **Charts**
 - Sales by Category (Bar Chart)
 - Sales by Segment (Donut Chart)
 - Profit by Region (Bar Chart)
 - Monthly Sales Trend (Line Chart)
 - Top 10 Products by Sales (Bar Chart)
- **Map Visualization**
 - Sales Distribution by State

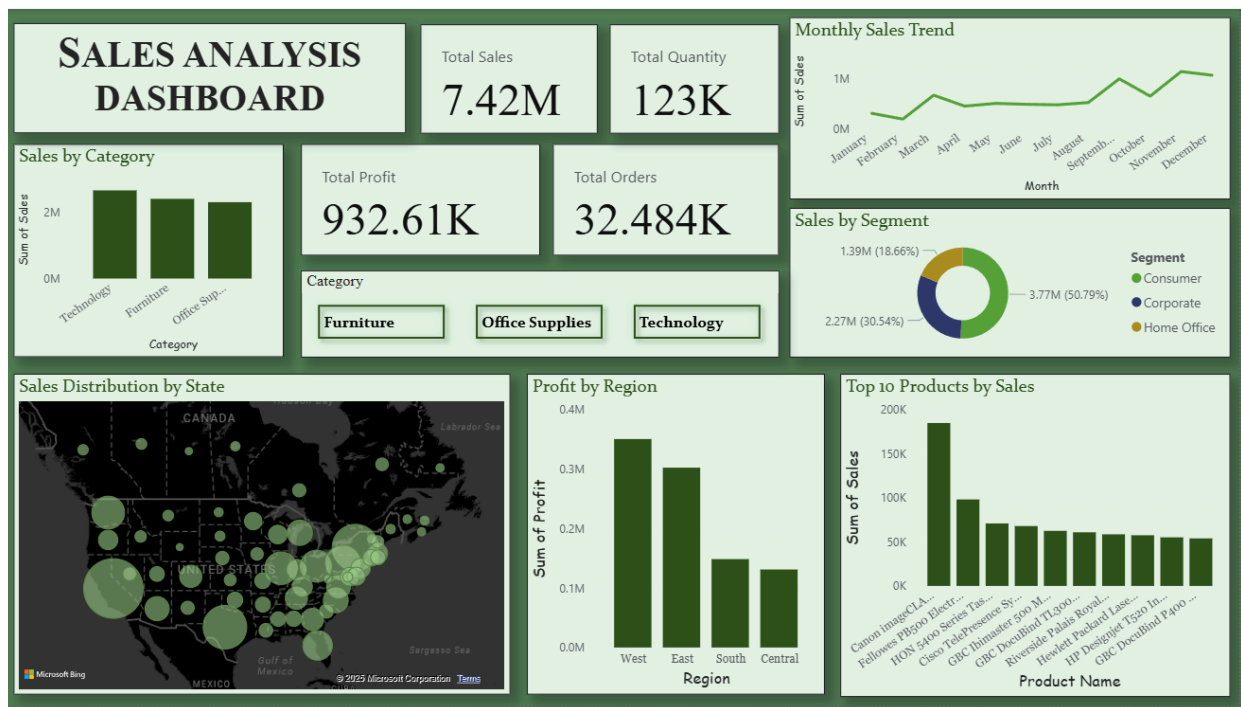
4.2 Key Insights

- ✓ Technology category generated highest revenue
- ✓ Consumer segment contributed majority of sales

- ✓ Western & Eastern regions earned highest profit
- ✓ November & December had strongest sales months
- ✓ Few products contributed significantly higher revenue

Additional insights:

- **Office Supplies** category, though lower in revenue, showed **steady demand**.
- **Corporate segment** had higher average order value, suggesting targeted upselling opportunities.
- **Regional disparities** highlight potential for expansion in underperforming areas.



Chapter 5: Business Impact & Inference

This dashboard helps businesses:

- Identify profitable regions
- Understand seasonal trends
- Focus on high-demand products
- Optimize marketing strategies
- Improve decision-making capability

Extended impact:

- **Inventory Management:** Align stock levels with seasonal demand.
- **Customer Targeting:** Personalize offers for high-value segments.
- **Strategic Planning:** Allocate resources to regions with growth potential.
- **Operational Efficiency:** Reduce costs by focusing on profitable categories.

Conclusion

This project successfully demonstrates how Power BI can transform raw sales data into powerful visual insights. The interactive dashboard enables better monitoring, strategic planning, and data-driven business decisions.

Beyond immediate insights, the project highlights the importance of **data governance, visualization best practices, and storytelling with data**. Overall, this project strengthens analytical thinking and real-world BI implementation skills.

References

- Power BI Documentation (Microsoft)
- Kaggle Sales Dataset (Sample)
- Business Intelligence Best Practices (Gartner Reports)

Appendix

- Sample Dashboard Screenshots
- Data Cleaning Scripts (Python)
- DAX Formulas Used in Power BI

